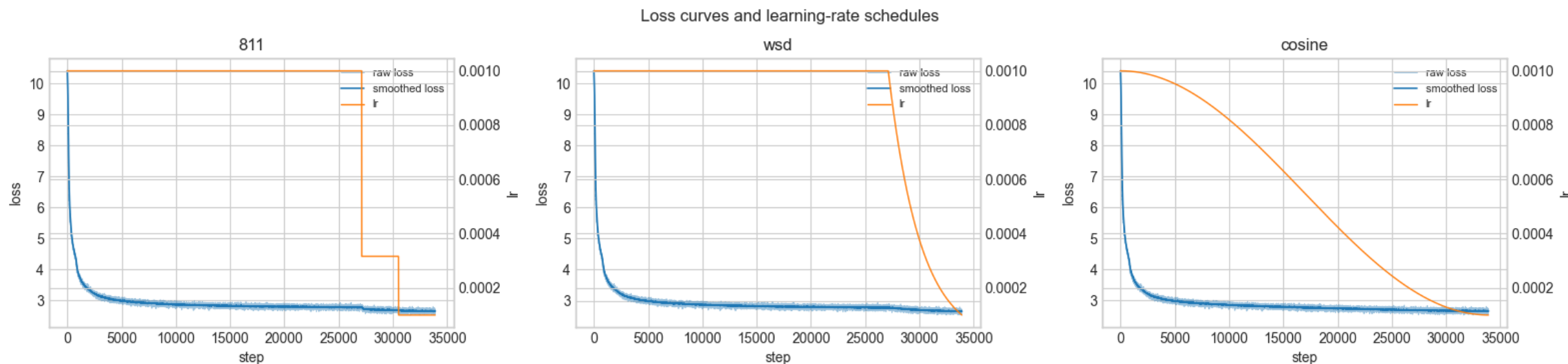




# Learning-Rate-Schedule-Aware Loss Curve Prediction

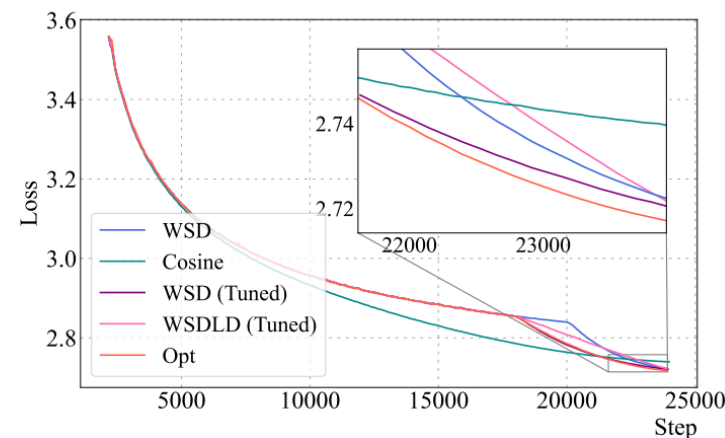
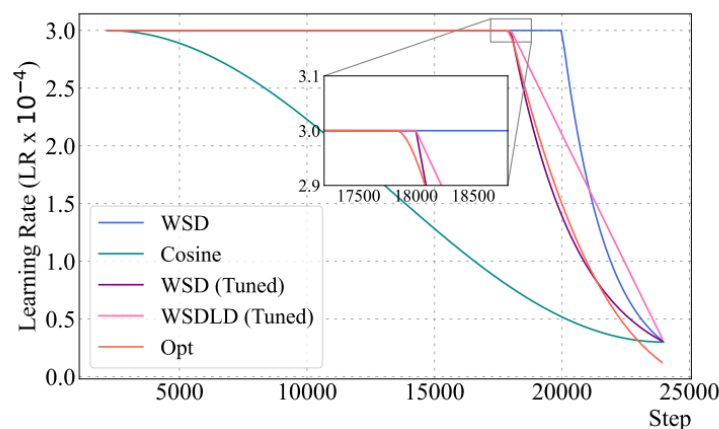
Zhirun Han 韩知润  
Student ID: 2401111536  
Graduate Student

# Loss Curve Prediction Across Learning-Rate Schedules



- **Goal:** predict the full pretraining loss curve under a new learning-rate schedule.
- **Required setting:** fit analytic models on the cosine schedule, then evaluate prediction performance on WSD.
- **Baselines reproduced:** One-Power Law, Tissue-style LR-annealing law, Multi-Power Law.
- **Our focus:** diagnose why cosine-to-WSD transfer fails, and test modified fitting strategies that reduce systematic residuals.

# Problem Background: Why schedule-aware loss prediction?



- Classical scaling laws mainly relate model/data scale to final validation loss.
- However, in LLM pretraining, the learning-rate schedule changes the entire trajectory of optimization.
- Cosine and WSD may have similar training budgets, but their loss curves can differ substantially during the stable and decay phases.
- Therefore, we study a harder but more useful problem:

$$L_t = L_{\Theta}(\eta_{\leq t})$$

where the loss at step (t) is predicted from the learning-rate history up to that step.

- **Key question:** Can a model fitted on one schedule, especially cosine, extrapolate to a structurally different schedule such as WSD?

# Task Definition and Main Questions

**Data:** cosine / WSD / 8-1-1 curves



Preprocess loss + LR



Fit analytic laws on cosine



Evaluate on WSD



Residual diagnosis + improved fitting

## Course-required experiment

- Fit models using the **cosine** learning-rate schedule.
- Evaluate the fitted models on the **WSD** schedule.
- Compare reproduced baselines and our modified methods.

## Research questions

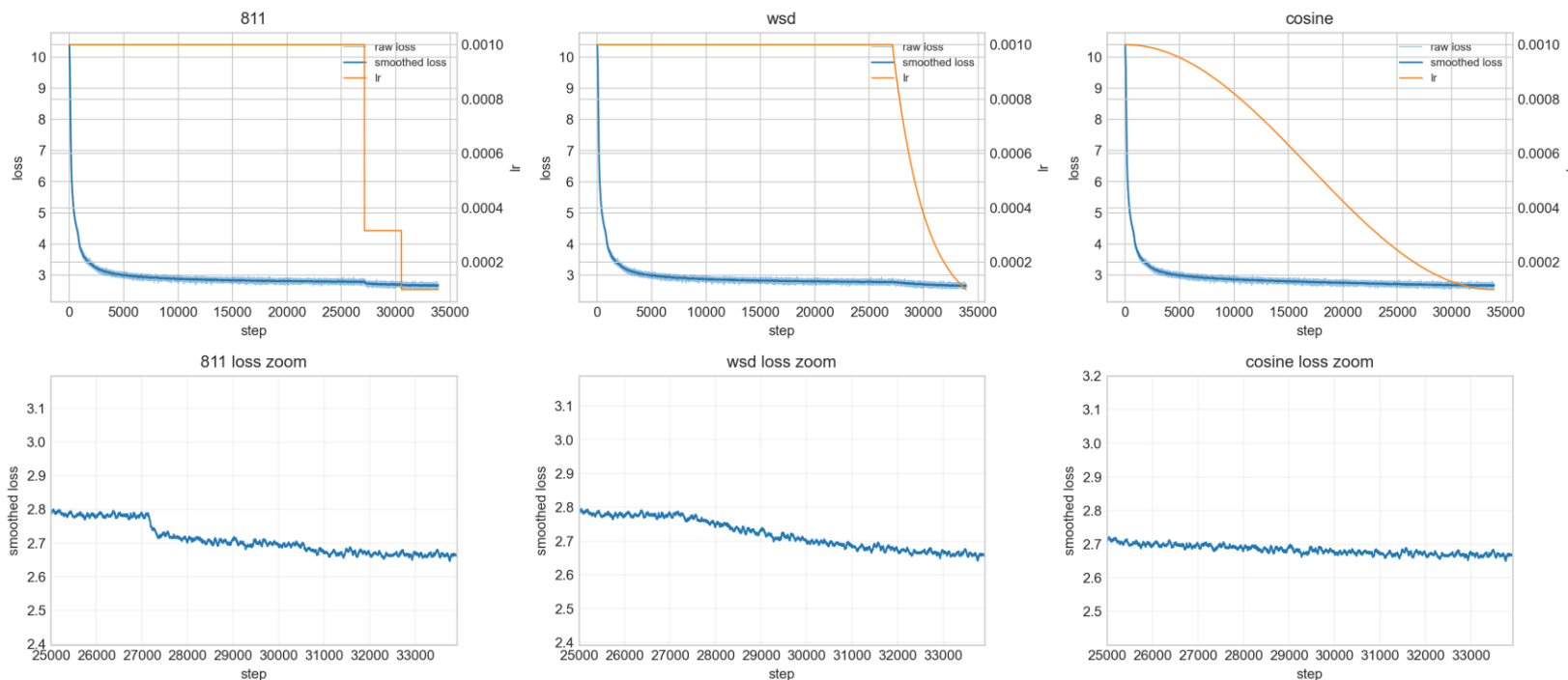
**Prediction accuracy:** Which analytic law predicts the WSD loss curve most accurately after fitting on cosine only?

**Failure mode:** If prediction is inaccurate, does the error mainly come from the WSD stable phase, the decay phase, or the final endpoint?

**Method development:** Can extra diagnostic schedules or decoupled fitting reduce systematic residuals without using WSD labels?

# Data and Preprocessing

Loss curves and learning-rate schedules



## Raw data

- Dataset file: gpt\_loss+lrs.pkl
- Three available schedules: wsd/cosine/811
- Each curve has 33,908 steps, from step 0 to step 33,907.
- Learning rate range:
  - max LR =  $(1.0 \times 10^{-3})$
  - min LR =  $(1.0 \times 10^{-4})$

## Preprocessing

- Missing step interpolation:
  - WSD: missing step 20815 filled.
  - Cosine: missing step 22493 filled.
- Missing values are filled by:
  - linear interpolation,
  - forward fill,
  - backward fill.
- Loss smoothing:
  - smoothing window = 80 steps.
  - smoothed loss is used for fitting and evaluation.
  - raw loss is kept for visualization

# Analytic Models for Schedule-Aware Loss Prediction

## 1. One-Power Law

- Treats cumulative learning rate as the effective training progress.
- Does not explicitly model LR decay or annealing.
- Used as the simplest ablation baseline.

$$L(t) = L_0 + AS_1(t)^{-\alpha}, \quad S_1(t) = \sum_{i=1}^t \eta_i$$

## 2. Tissue-style LR-Annealing Law

- $S_1$ : forward area, measuring accumulated optimization progress.
- $S_2$ : annealing momentum, measuring historical LR decrease.
- We use ( $\lambda = 0.999$ ) in later decoupled experiments.

$$L(t) = L_0 + AS_1(t)^{-\alpha} - CS_2(t)$$

$$S_2(t) = \sum_{i=1}^t \sum_{k=1}^i (\eta_{k-1} - \eta_k) \lambda^{i-k}$$

## 3. Multi-Power Law

- Models LR-drop-induced loss reduction with a power-law saturation term.
- More expressive than Tissue-style law.
- Potentially more accurate, but also has more parameters and stronger risk of parameter entanglement.

$$L(t) = L_0 + AS_1(t)^{-\alpha} - LD(t)$$

$$LD(t) = \sum_{k=2}^t B(\eta_{k-1} - \eta_k) \left[ 1 - (C\eta_k^{-\gamma} S_k(t) + 1)^{-\beta} \right]$$

# Experimental Setup: Fit on Cosine, Evaluate on WSD

## Main required setting

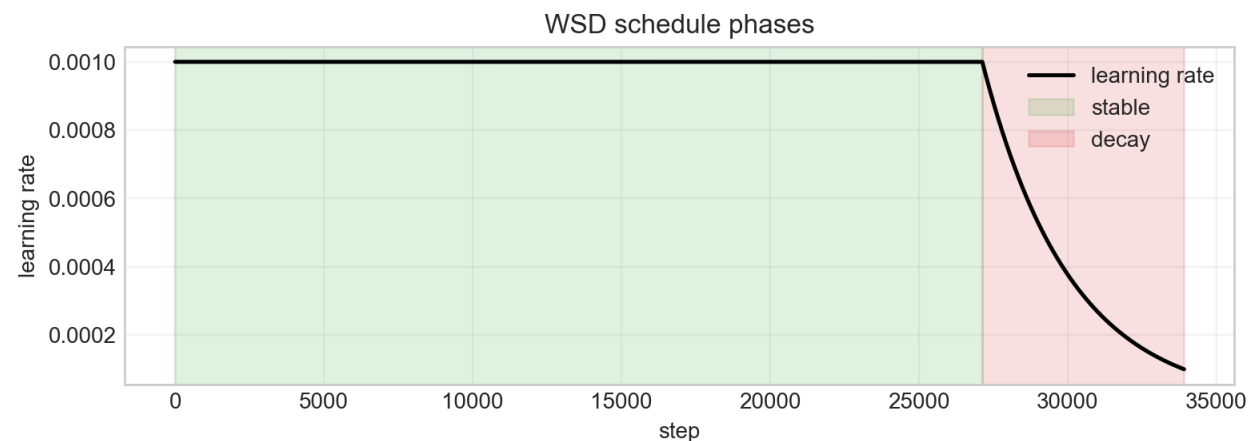
- Training schedule: cosine
- Test schedule: WSD
- WSD loss labels are not used during baseline model fitting.

## Fitting setup

- Model fitting starts from step 1,000.
- Fitting points are sampled every 500 steps.
- The maximum number of optimization steps for MPL is 2,000.

## Evaluation setup

- Predictions are evaluated every 100 steps.
- Evaluation metrics include:
  - Mean Absolute Error (MAE)
  - Root Mean Squared Error (RMSE)
  - Mean Absolute Percentage Error (MAPE)
  - Worst Absolute Percentage Error (Worst APE)
  - Coefficient of Determination ( $R^2$ )
  - Final Absolute Error

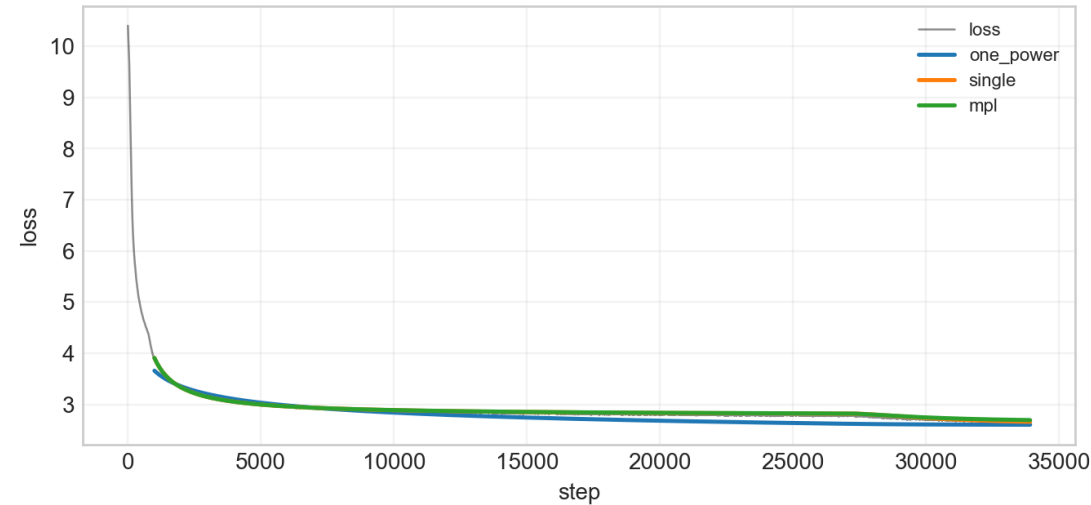


- **Stable phase:** steps 0–27125, total 27,126 steps.
- **Decay phase:** steps 27126–33907, total 6,782 steps.
- This split is used later to diagnose whether residuals come from the long stable plateau or from the final decay stage.

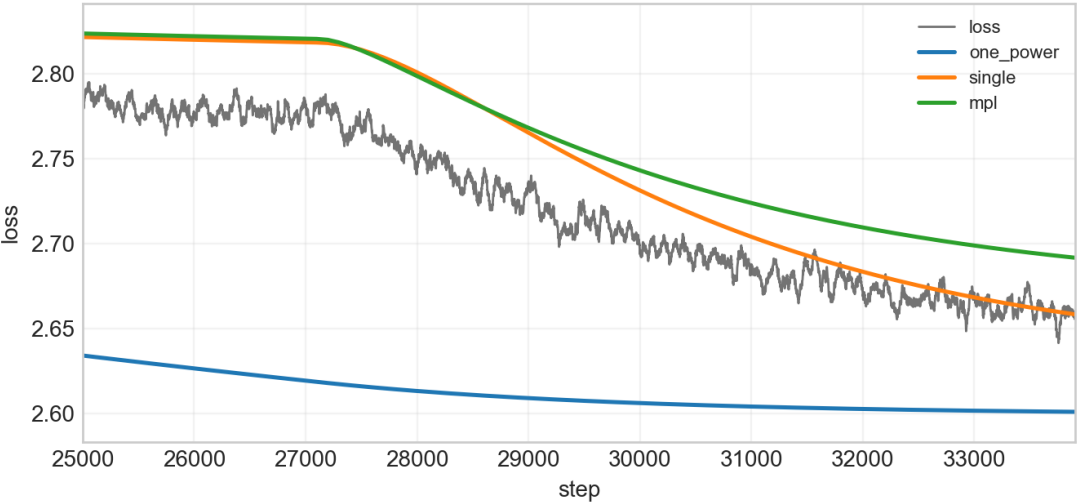
# Baseline Reproduction: Cosine $\rightarrow$ WSD Prediction

WSD prediction after fitting on cosine only

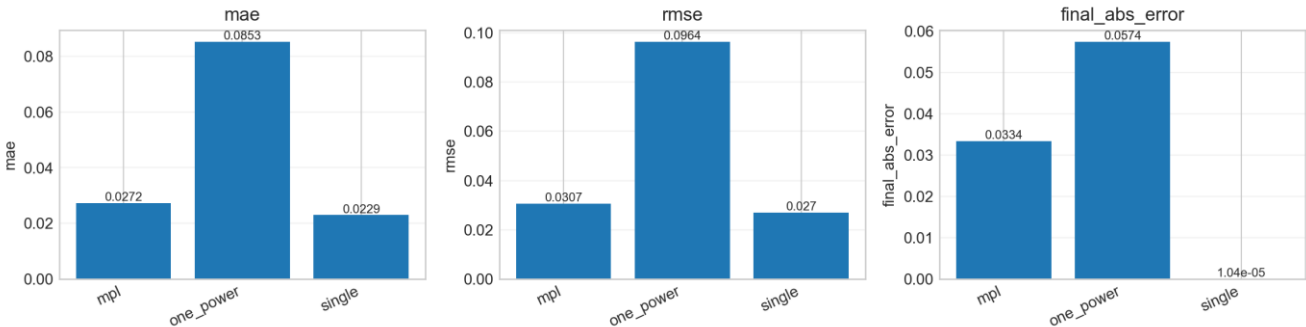
wsd



wsd prediction zoom



WSD prediction metrics after fitting on cosine only



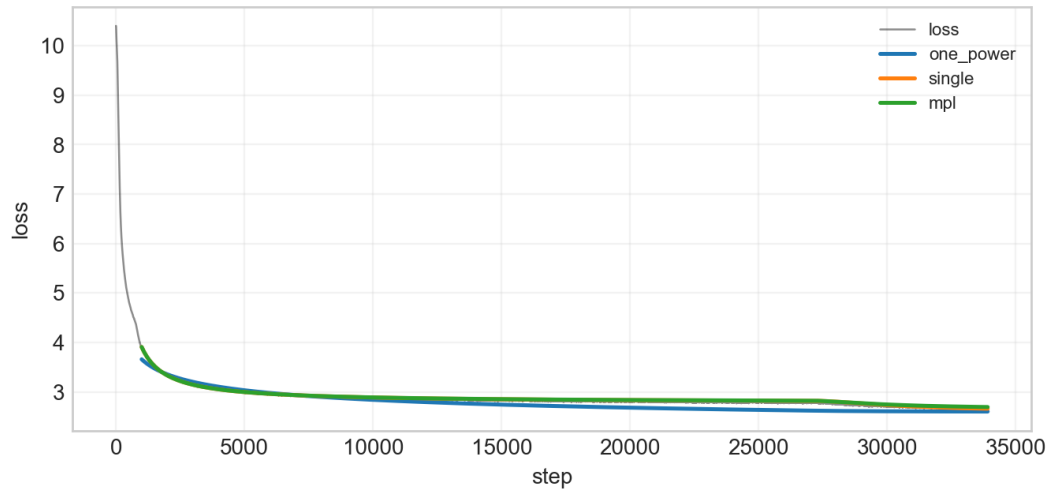
| Method           | MAE    | RMSE   | Final Abs. Error        |
|------------------|--------|--------|-------------------------|
| One-Power Law    | 0.0853 | 0.0964 | 0.0574                  |
| Tissue-style Law | 0.0229 | 0.0270 | $(1.04 \times 10^{-5})$ |
| Multi-Power Law  | 0.0272 | 0.0307 | 0.0334                  |



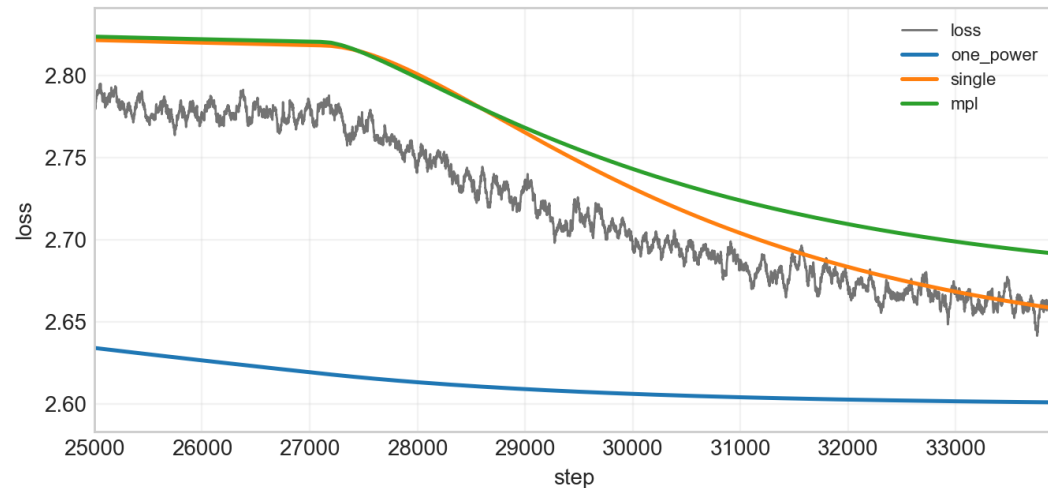
# Baseline Reproduction: Cosine $\rightarrow$ WSD Prediction

WSD prediction after fitting on cosine only

wsd



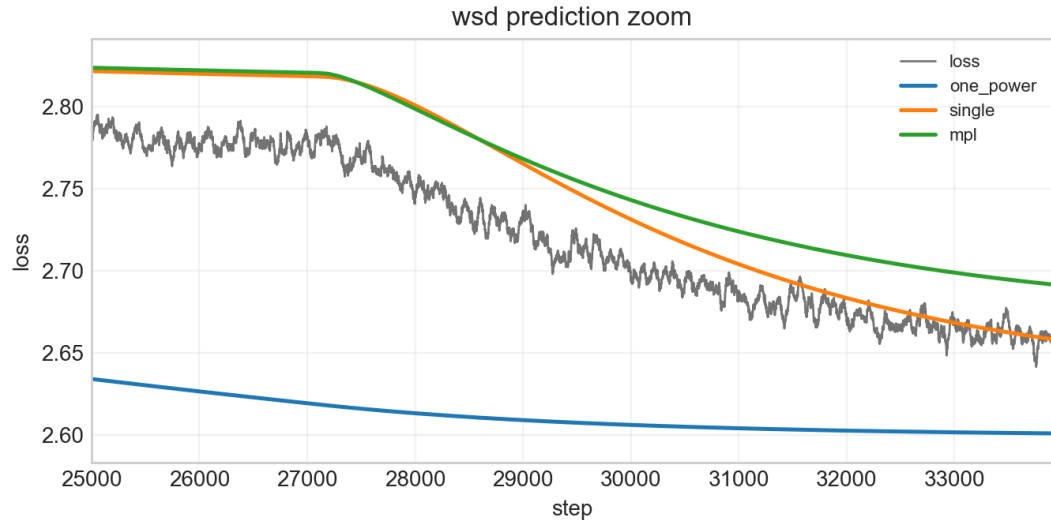
wsd prediction zoom



## Observations

- One-Power Law performs clearly worse, showing that cumulative LR area  $S_1$  alone is insufficient.
- Tissue-style Law gives the lowest global MAE/RMSE under cosine-only fitting.
- Tissue-style Law also has an extremely small final error.
- MPL is more expressive, but under cosine-only fitting its WSD prediction is not better than Tissue-style Law.

# Residual Diagnosis: Where Does the Prediction Fail?

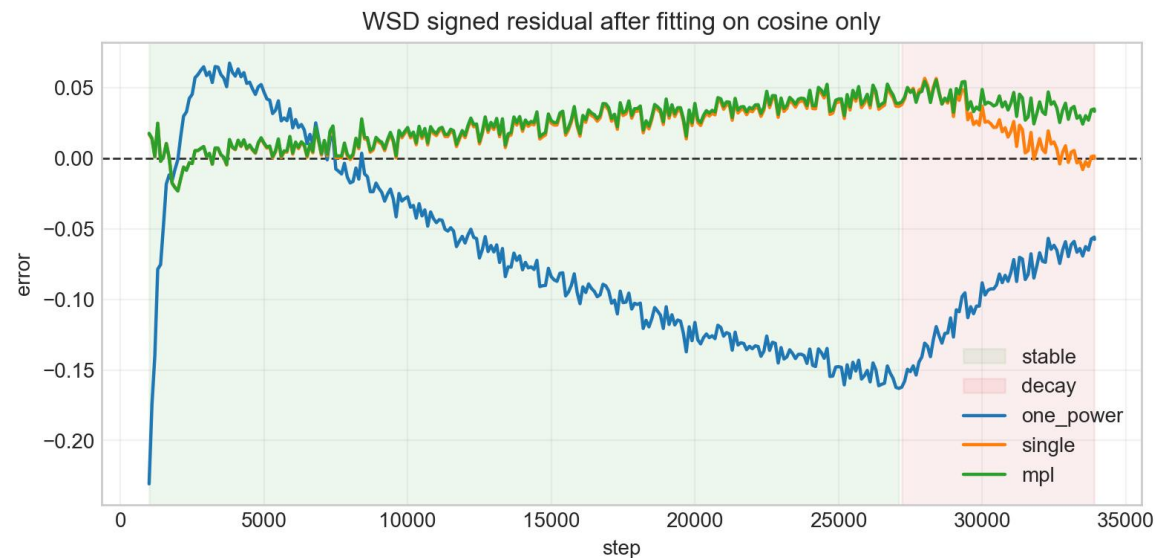


## Stable phase

- Tissue-style Law and MPL show similar positive residuals.
- The residual grows almost linearly during the constant-LR stage.
- Both methods systematically overestimate WSD loss.

## Decay phase

- Residuals do not drop immediately after LR decay begins.
- After a short delay, they decrease as the decay effect appears.
- The decrease becomes faster in the later decay stage.



# Diagnostic Experiment: Does WSD Need Stable-Phase Information?



## Motivation from residual diagnosis

- In cosine-only fitting, both Tissue-style Law and MPL overestimate the WSD loss during the stable phase.
- The residual grows almost linearly before LR decay starts.
- This suggests that the fitted models may not have learned the correct base dynamics under a long constant-LR plateau.

## Diagnostic idea

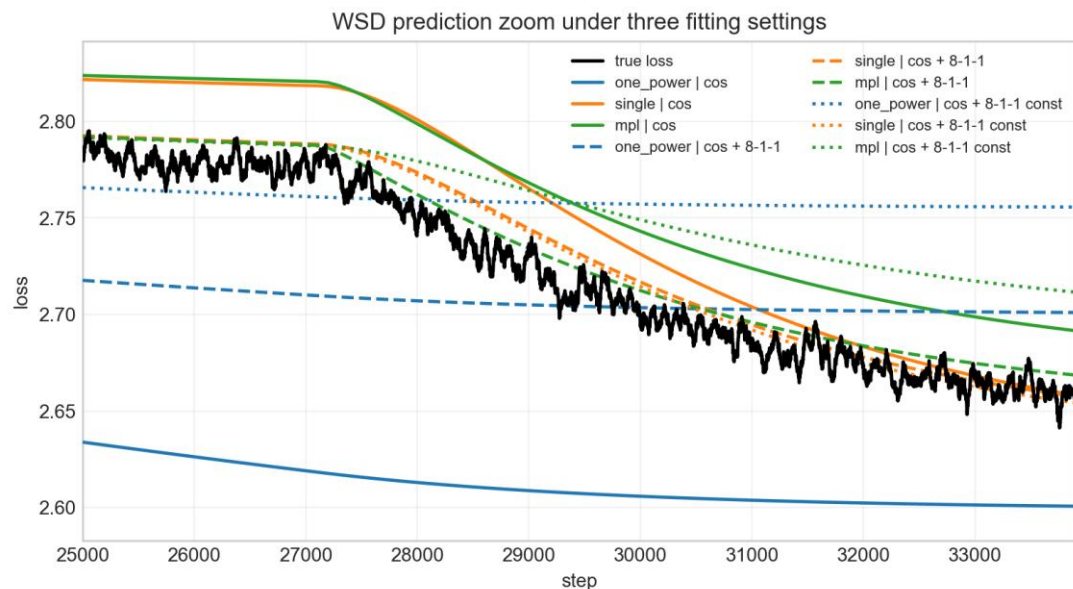
We introduce the 8-1-1 schedule as an auxiliary diagnostic curve.

- The 8-1-1 schedule contains a long high-LR segment, similar to the stable phase of WSD.
- It also contains explicit step-wise LR drops, which provide information about LR-decay-induced loss reduction.
- This allows us to separately test whether the WSD prediction error comes from:
  - missing stable-phase dynamics,
  - missing decay-response information,
  - or both.

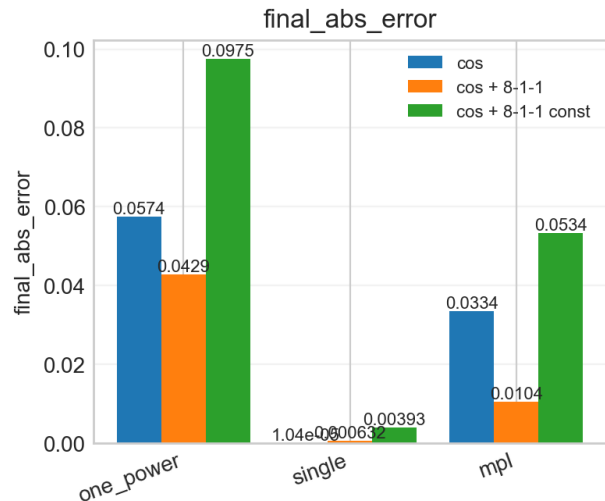
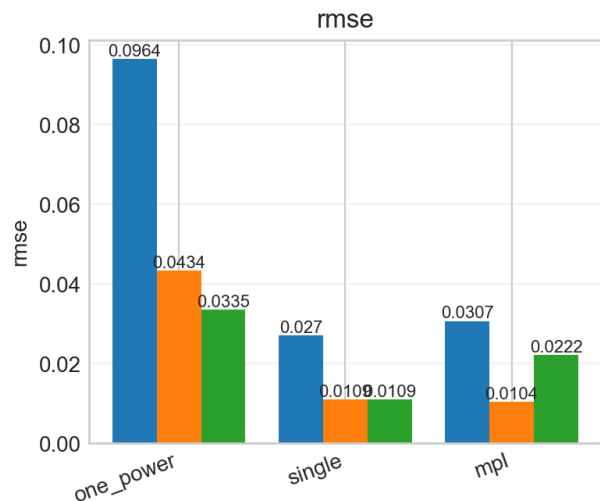
## Three fitting settings

| Setting              | Fitting data                       | Purpose                                    |
|----------------------|------------------------------------|--------------------------------------------|
| Cosine only          | cosine                             | required baseline                          |
| Cosine + 8-1-1       | cosine + full 8-1-1                | add both stable and step-decay information |
| Cosine + 8-1-1 const | cosine + constant segment of 8-1-1 | add stable-phase information only          |

# Diagnostic Experiment: Does WSD Need Stable-Phase Information?



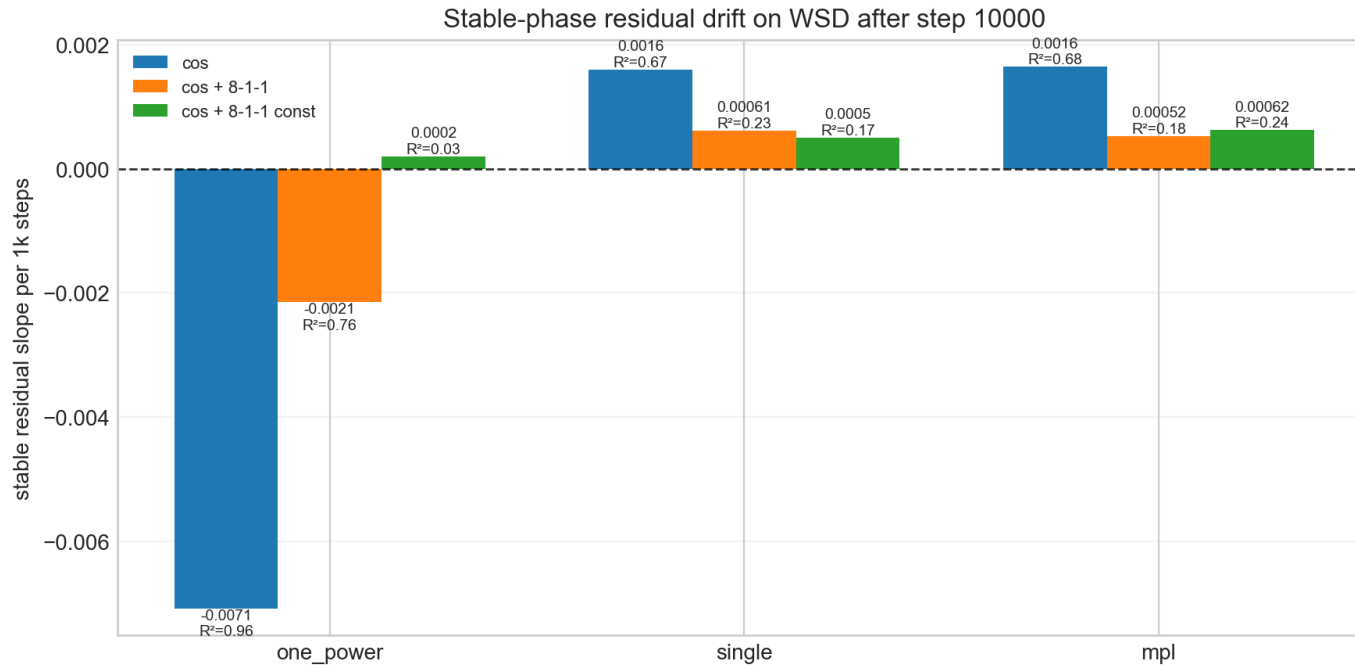
WSD metrics under three fitting settings



## Key observations

- Adding 8-1-1 or 8-1-1 const greatly reduces stable-phase bias for both Single and MPL, showing that long constant-LR dynamics are the missing information in cosine-only fitting.
- For Single, 8-1-1 and 8-1-1 const give nearly identical WSD predictions (MAE 0.0089 vs. 0.0090), indicating that explicit decay data provides little additional benefit.
- For MPL, full 8-1-1 improves the early WSD decay fit, suggesting that MPL benefits from explicit decay examples. However, MPL trained with only 8-1-1 const performs poorly in the decay phase, implying that its decay-correction term requires LR-drop data to be identified.

# Quantifying the Stable-Phase Drift



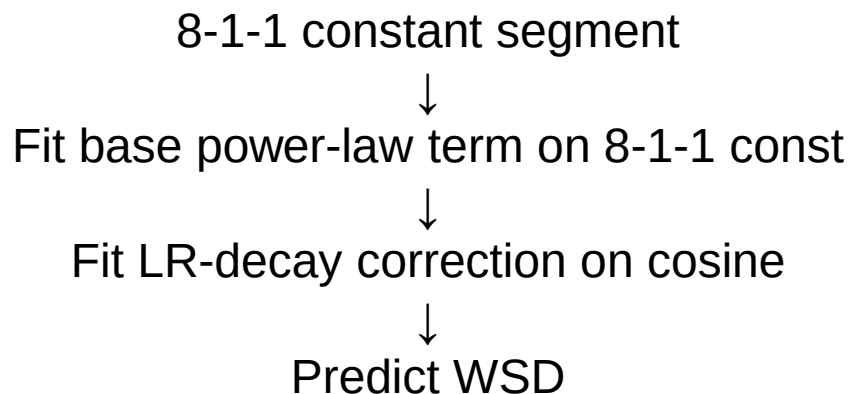
## Residual slope definition

- In the WSD stable phase, residuals increase almost linearly.
- We fit a simple line after step 10,000:

$$e_t \approx a + b \cdot \frac{t - t_0}{1000}$$

b measures residual drift per 1,000 steps.

- For Single and MPL, cosine-only fitting has the largest positive stable-phase drift.
- Adding either full 8-1-1 or only 8-1-1 const reduces this drift by about 2.5–3×.
- The improvement is already visible with 8-1-1 const, confirming that stable-phase calibration is the key factor.



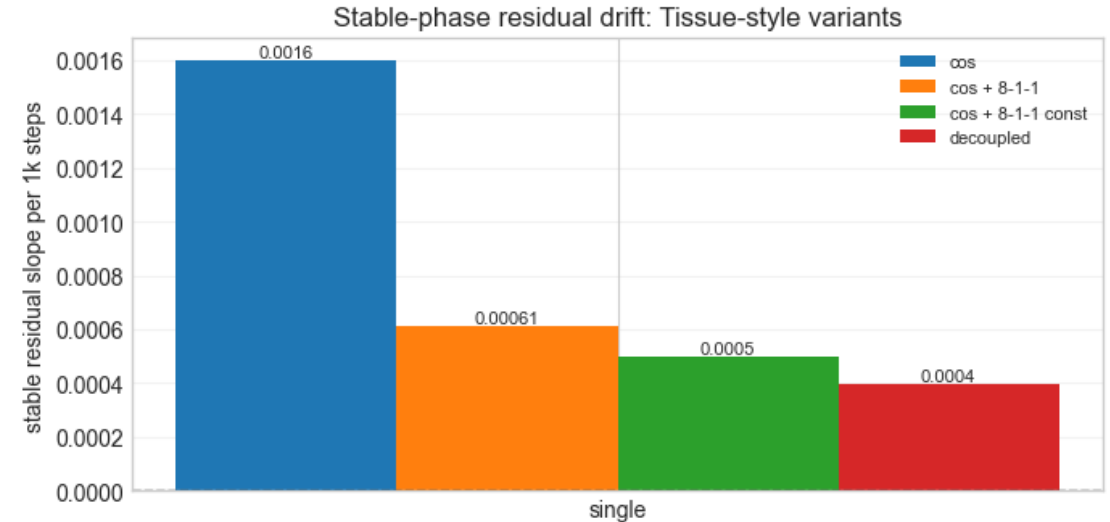
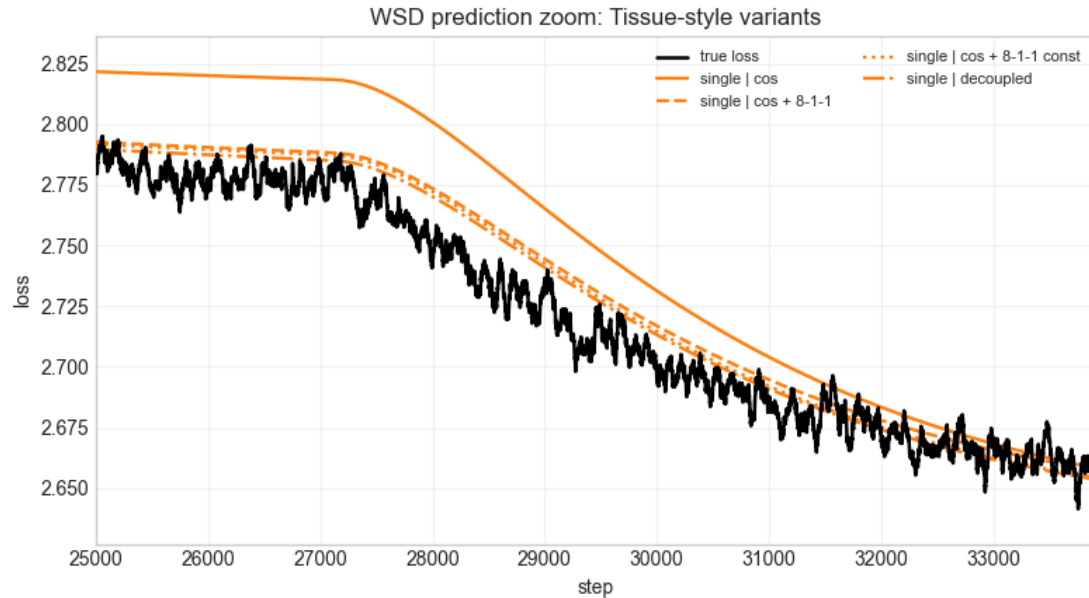
## Motivation

- Previous residual analysis suggests that the error has two sources:
  - base learning dynamics under a long constant-LR phase;
  - additional loss reduction after LR decay begins.
- Joint fitting may entangle these two effects.
- Therefore, we fit them in two separate stages.

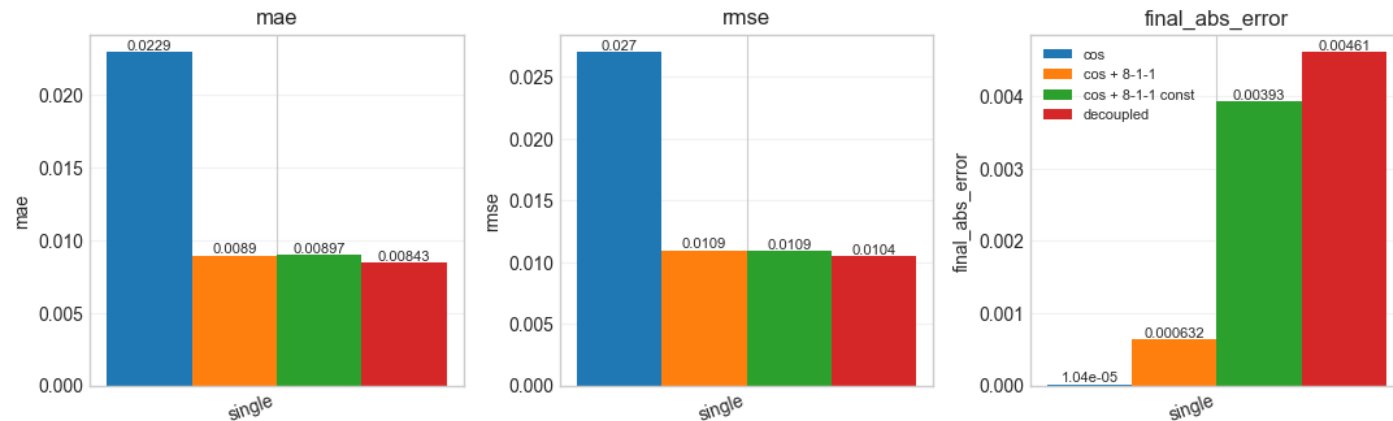
## Training data used

- Base term: fitted using the constant segment of 8-1-1.
- Decay correction: fitted using cosine
- WSD loss labels are still not used during fitting.

# Decoupled Single Law: Slightly Better than Joint Cosine + 811 Const

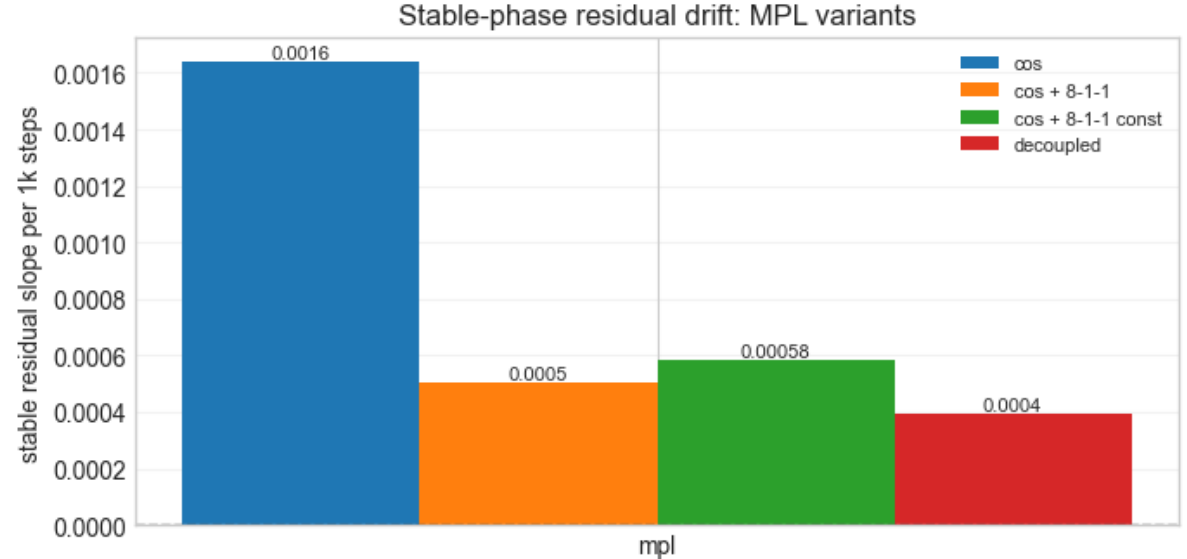
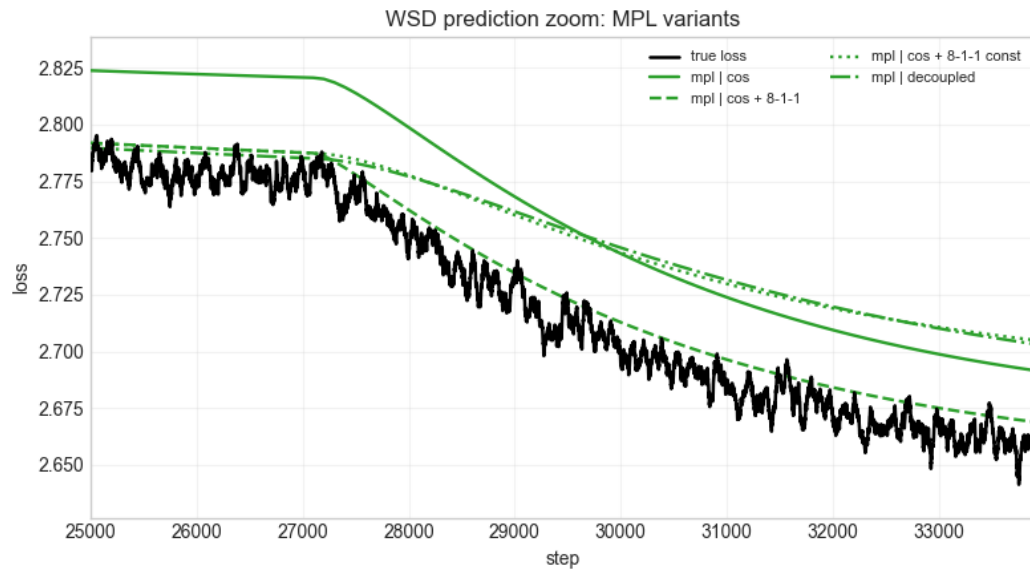


WSD metrics: Tissue-style variants

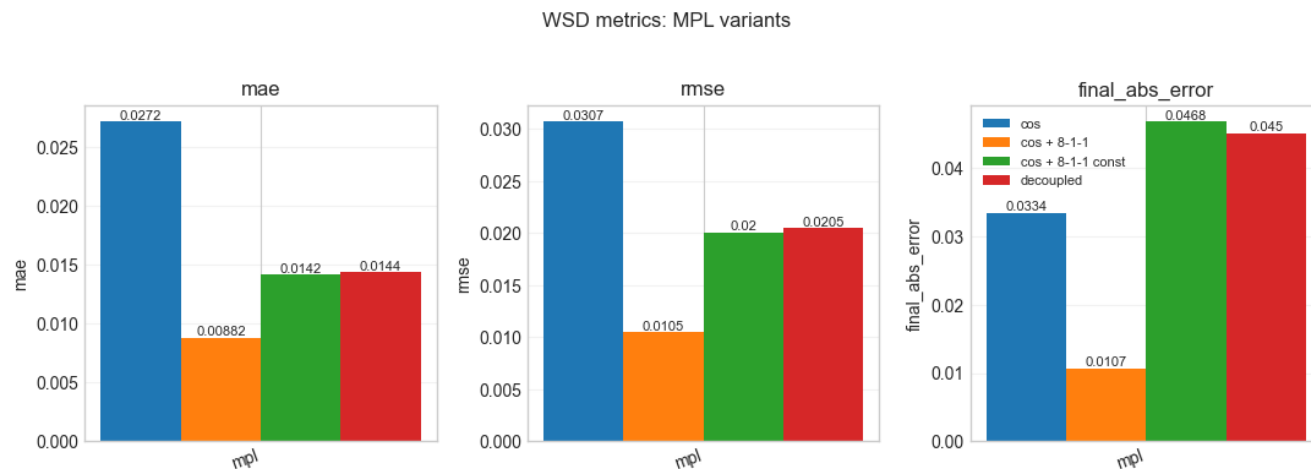


- Decoupled Single is very close to the joint cosine + 8-1-1 const result. This supports the idea that the main useful information is the long constant-LR segment.
- Separating base fitting from correction fitting slightly reduces stable-phase drift.
- However, the improvement is not dramatic, meaning joint fitting with 8-1-1 const already captures most of the gain.

# Decoupled MPL: Similar to Joint Cosine + 8-1-1 Const, But Still Below Full 811



- Decoupled MPL slightly improves over joint cosine + 8-1-1 const.
- Stable-phase drift is reduced, mainly because the base term is calibrated on the constant-LR segment.
- However, without full 8-1-1 decay information, the MPL decay correction remains underdetermined.
- This is why Decoupled MPL is still clearly worse than joint MPL trained with full 8-1-1.
- Its final error is also large, showing that stable-phase calibration alone is not sufficient for MPL's flexible decay term.





# Additional Trial: Schedule-Feature Residual Correction

## Motivation

- The residual analysis suggests that part of the error has a smooth, schedule-dependent structure.
- We therefore test whether a lightweight residual model can correct the analytic prediction.

## Method

- Start from the fitted Single / Tissue-style prediction.
- Add a ridge regression correction:

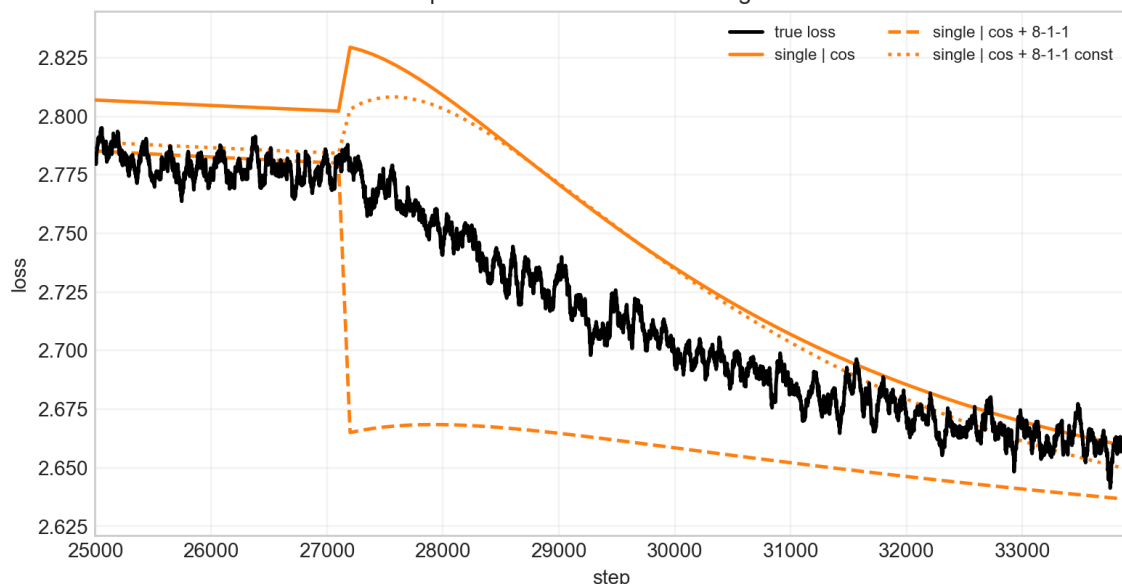
$$\hat{L}_{\text{single}}(t) + w^\top x_t$$

- Ridge regularization  $\alpha = 10^{-2}$

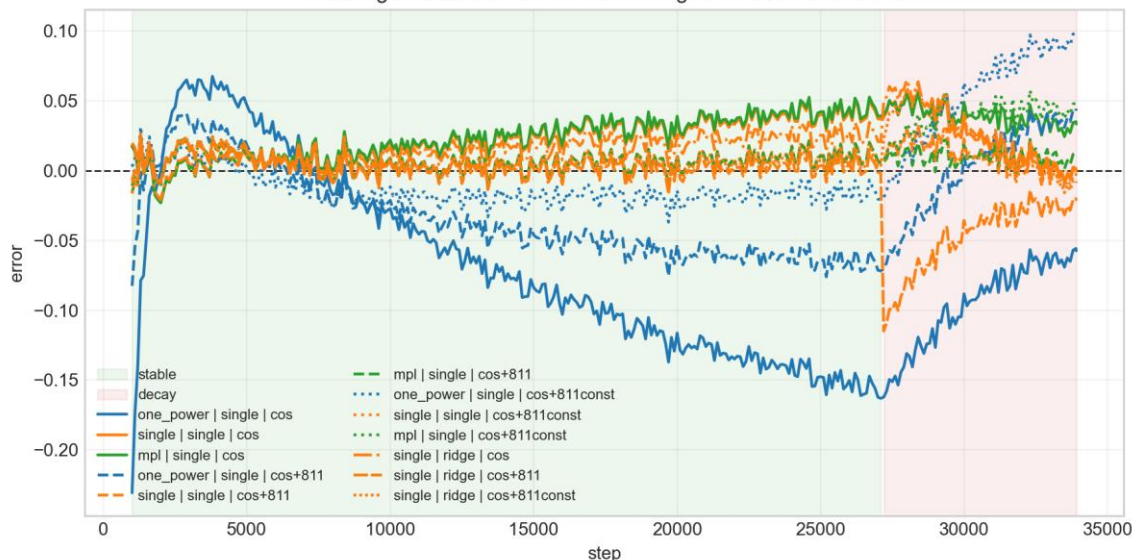
| Feature name         | Meaning                                        |
|----------------------|------------------------------------------------|
| t_frac               | Normalized training step: $t/t_{\max}$         |
| s1_frac              | Normalized cumulative LR area: $S_1(t)/S_1(T)$ |
| lr_frac              | Current LR divided by maximum LR               |
| remaining_s1_frac    | Remaining LR area: $(S_1(T) - S_1(t))/S_1(T)$  |
| s2_frac              | Normalized LR-annealing momentum feature       |
| cum_drop_frac        | Cumulative positive LR drop up to step $t$     |
| current_drop_frac    | Current positive LR drop at step $t$           |
| time_since_drop_frac | Normalized time since the most recent LR drop  |

# Additional Trial: Schedule-Feature Residual Correction

WSD prediction zoom: Tissue + ridge residual



WSD signed residual: Tissue law vs ridge-corrected Tissue law



## Result

- The ridge residual correction only slightly improves over the One-Power Law baseline.
- However, it is still worse than the original Tissue-style model fitted on cosine only.
- Therefore, adding a flexible residual correction on top of a weak base model does not recover the schedule-aware structure captured by the analytic Tissue-style ( $S_2$ ) term.

## Interpretation

- The schedule-derived features contain some useful information, but they do not replace an explicitly designed LR-annealing term.
- The correction is fitted as a generic linear residual model, so it mainly captures smooth trends rather than the true decay-response mechanism.
- Its limited improvement suggests that the key issue is not simply adding more hand-crafted schedule features.

# Overall Findings and Takeaways

| Method / setting                | WSD<br>MAE | Main behavior                                                      |
|---------------------------------|------------|--------------------------------------------------------------------|
| Single, cosine only             | 0.0229     | strong stable-phase overestimation; final error partly cancelled   |
| MPL, cosine only                | 0.0272     | strong stable-phase overestimation; decay response underidentified |
| Single, cosine + 8-1-1<br>const | 0.0090     | stable phase largely fixed                                         |
| Single, decoupled               | 0.00843    | slightly better than joint <code>cos + 8-1-1 const</code>          |
| MPL, cosine + 8-1-1<br>const    | 0.0153     | stable improves, but decay remains poor                            |
| MPL, decoupled                  | 0.01440    | slightly better than joint <code>cos + 8-1-1 const</code>          |
| MPL, cosine + full 8-1-1        | 0.0087     | best MPL joint fit; explicit decay information helps               |

- **Cosine-only fitting is under-identified for WSD.**  
Both Single and MPL overestimate the loss during the long stable phase.
- **Stable-phase calibration is the dominant factor.**  
Adding only the constant segment of 8-1-1 already removes most of the residual drift.
- **More flexible models need more decay information.**  
MPL benefits from full 8-1-1 decay data, but its decay correction is also more sensitive to decay-shape mismatch.

- Reference
  - Tissue, H., Wang, V., and Wang, L. (2024). *Scaling Law with Learning Rate Annealing*.
  - Luo, K., Wen, H., Hu, S., Sun, Z., Liu, Z., Sun, M., Lyu, K., and Chen, W. (2024). *A Multi-Power Law for Loss Curve Prediction Across Learning Rate Schedules*.
  - Li, B., Chen, F., Huang, Z., Wang, L., and Wu, L. (2025). *Functional Scaling Laws in Kernel Regression: Loss Dynamics and Learning Rate Schedules*.
  - Zhang, H., Wen, K., and Ma, T. (2026). *Configuration-to-Performance Scaling Law with Neural Ansatz*
- This is an individual project. Zhirun Han: preprocessing, baseline reproduction, method development, result analysis, slides and code organization.
- Code and reproducibility
  - GitHub: <https://github.com/Hubuu/loss-curve-prediction.git>
  - Main notebook: code/loss\_curve\_prediction\_final.ipynb
  - Data path: loss curves/gpt\_loss+lrs.pkl
  - Run instructions:
    1. Install dependencies: pip install -r requirements.txt
    2. Run: code/loss\_curve\_prediction\_final.ipynb
  - Outputs: code/results/figures/ and code/results/tables/